DeepWatch: A Unified Multimodal Platform for Deepfake Forensic Analysis and Generation-Method Attribution

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Abstract

Deepfake media — synthetic faces, voices, and videos produced by generative models — is increasingly used for fraud, impersonation, and fabricated evidence. Published research addresses the problem in fragments: detectors output opaque scores, attribution methods are restricted to facial forgeries, and speech anti-spoofing is evaluated in isolation from video. This proposal analyses four IEEE/CVF publications, identifies their documented limitations, and proposes **DeepWatch**, a forensic platform that integrates detection, generation-method attribution, explainable evidence, and integrity-preserving forensic reporting into a single pipeline. The expected contribution is an evaluated, reproducible system whose components are measured against the base papers' published benchmarks.

Keywords *— Deepfake detection, digital forensics, attribution, audio–video fusion, explainable AI, evidence integrity.*

1. Introduction and Problem Statement

Generative AI enables non-experts to fabricate convincing faces, voices, and video within minutes. Reported harms include voice-clone fraud calls, fabricated video of private individuals, and unverifiable digital "evidence." A victim or small investigative unit currently has no accessible means to demonstrate that media is manipulated: existing detectors return a bare probability, trained on controlled datasets, and are not designed to produce evidence usable by third parties. The problem addressed here is therefore twofold: (a) determining whether media is manipulated, and (b) documenting that determination in a forensically defensible manner.

2. Literature Review and Research Gap

Four publications in verified IEEE venues were selected as the technical foundation. Their methods, strengths, and limitations are compared below.

Base paper	Method / contribution	Documented limitation
UCF, Yan et al., ICCV 2023 [1]	Disentangles "common" forgery features from method-specific ones for generalizable detection; 0.996 AUC in-dataset.	Average AUC falls to 0.828 on unseen datasets; requires per-method forgery labels; produces no explanation.
Up-sampling artifacts, Tan et al., CVPR 2024 [2]	Detects fakes via artifacts left by CNN-generator up-sampling layers.	Targets CNN-generator artifacts; diffusion-based generators do not share this artifact class.
OW-DFA / CPL, Sun et al., ICCV 2023 [3]	Open-world attribution of unseen forgery methods via contrastive pseudo learning.	Evaluated on facial forgeries only; reduced accuracy on unseen methods; the attribution subfield is early-stage (≈27–29 citations).
ASVspoof 2021, Liu et al., IEEE/ACM TASLP 2023 [4]	Community benchmark for spoofed and deepfake **speech** detection.	Audio-only evaluation; not combined with visual analysis, though video calls contain both channels.

Research gap. Detection, attribution, and speech anti-spoofing progress as separate tracks; the reviewed literature contains no integrated system that performs all three and emits explainable, integrity-protected evidence. Within the scope of this literature review — which is limited to the sources surveyed and does not claim to be exhaustive — this combination appears unaddressed.

3. Research Objectives and Proposed Methodology

DeepWatch is proposed to close the identified gaps through five components. These are **proposed features**, not verified results.

- **Diffusion-aware detection.** Extend artifact-based detection beyond CNN-generator fingerprints [2] toward diffusion-era generators, using frequency-domain and noise-pattern analysis.
- **Audio–video fusion.** Jointly analyse the visual channel (after [1]), the speech channel (after [4]), and lip–speech synchronisation, so that media with consistent visuals but synthetic audio remains detectable.
- **Generation-method attribution.** Extend the open-world approach of [3] to estimate the most probable generation method, and report an explicit "unseen method" outcome where confidence is insufficient.
- **Explainable evidence.** Produce region-level indicators (heatmaps, artifact localisations) with plain-language descriptions. These outputs are **indicators supporting an assessment, not definitive proof** of manipulation or authenticity.
- **Forensic report generation.** Emit a structured report containing all scores, visualisations, hashes, and method metadata, suitable for submission to investigators or platform moderators.

4. Digital Forensic Framework and Evaluation Plan

Evidence integrity. On ingestion, each submitted file is hashed with SHA-256; the hash, timestamp, and source metadata are logged before analysis. The original media is preserved unmodified; all processing operates on copies. Every analysis step, model version, and parameter is recorded in an append-only analysis log to support traceability and independent re-examination. A report hash may optionally be anchored on a public test blockchain as a tamper-evidence measure — noting that blockchain anchoring proves a report's integrity after creation, not the truth of its contents.

Datasets and baselines. Evaluation will use the public benchmarks established by the base literature: FaceForensics++ with cross-dataset evaluation on Celeb-DF and DFDC for visual detection [1], the OW-DFA benchmark for attribution [3], and ASVspoof 2021 partitions for speech [4]. Detection will be compared against the Xception baseline reported by [1] and the up-sampling method [2]; attribution against CPL [3]. Ablations will isolate the fusion component's contribution.

Metrics. AUC and EER for detection; top-k attribution accuracy and open-set metrics for attribution; calibration error for confidence outputs. False-positive and false-negative rates will be reported separately — a false negative shields a forgery, while a false positive falsely accuses authentic media; both carry real-world cost and will be quantified rather than suppressed into a single accuracy figure.

Reproducibility. Code, configurations, model hashes, and evaluation scripts will be released so third parties can rerun the evaluation and obtain identical results.

5. Expected Contributions, Challenges, and Conclusion

Expected contributions. (a) An integrated detect–attribute–explain–report pipeline as an evaluated prototype; (b) an audio–video fusion method measured on realistic, compression-degraded media; (c) a reproducible evaluation against published baselines. These are planned contributions pending experimental validation.

Challenges and limitations. Cross-dataset generalization remains the field's hardest open problem; improvement targets must be stated modestly. Social-media compression destroys low-level artifacts. Diffusion-era evaluation datasets are less standardized. Detection outputs are probabilistic indicators, not proof; human review remains necessary for consequential decisions.

Conclusion. Deepfake forensics is a current, under-served problem. The four base papers provide validated components but leave documented gaps — degraded cross-dataset accuracy, blindness to newer generators, face-only attribution, and unimodal audio evaluation. DeepWatch is proposed to close these gaps as a unified, evidence-producing platform, with all claimed improvements to be verified experimentally against published benchmarks.

References

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